Data Preparation

Date	21 October 2026
Team ID	EN25CS3010849
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	5 Marks

Data preparation refers to the process of cleaning, standardizing, transforming, and organizing data so that it is suitable for analysis and reporting. In a Snowflake project, data preparation involves working with the ingested RAW data, identifying data quality issues, creating STAGING tables, applying required transformations, and preparing reliable data for the CURATED and MART layers.

Activity 1: Identifying and Handling Null Values

- **Objective:** Identify NULL or missing values present in the athlete and medal datasets, analyze affected columns (such as Age, Height, Weight, or Medal), and apply appropriate SQL queries to handle or impute them.

SQL Query:

SQL

```
-- Check for NULL or missing values in athlete physical attributes and medals
SELECT
  COUNT(*) AS Total_Rows,
  SUM(CASE WHEN Age IS NULL THEN 1 ELSE 0 END) AS Missing_Age,
  SUM(CASE WHEN Height IS NULL THEN 1 ELSE 0 END) AS Missing_Height,
  SUM(CASE WHEN Weight IS NULL THEN 1 ELSE 0 END) AS Missing_Weight,
  SUM(CASE WHEN Medal IS NULL THEN 1 ELSE 0 END) AS Missing_Medals
FROM OLYMPIC_PERFORMANCE_DB.RAW.ATHLETE_RAW;
```

Output / Result:

OTAL_ROWS	ISSING_AGE	ISSING_HEIGHT	ISSING_WEIGHT	ISSING_MEDALS
71116	474	0171	2875	31333

- **Handling Strategy:** Missing values in the Medal column indicate non-medal-winning participation records, which are valid and retained as NULL. Missing physical attributes (Height, Weight, and Age) are filtered out during specific attribute-based correlation analyses or handled using sport-specific medians.

Activity 2: Duplicate Record Handling

- Objective: Identify and analyze duplicate rows in the raw Olympic tables to ensure that redundant entries do not skew statistical counts or medal tallies.

SQL Query:

SQL

-- Identify duplicate entries based on Athlete ID, Games, Event, and Medal

SELECT

ID, Name, Games, Event, Medal,

COUNT(*) as Duplicate_Count

FROM OLYMPIC_PERFORMANCE_DB.RAW.ATHLETE_RAW

GROUP BY ID, Name, Games, Event, Medal

HAVING COUNT(*) > 1;

Output / Result:

D	IAME	AMES	VENT	MEDAL	UPLICATE_COUNT
521	Michael Fred Phelps, II	008 Summer	wimming Men's 200 metres Butterfly	old	
532	Usain St. Leo Bolt	016 Summer	thletics Men's 100 metres	old	

- Handling Strategy: Duplicate rows resulting from multi-team event entries or overlapping raw logs are cleaned using Snowflake's deduplication logic to maintain distinct event records.

Activity 3: Data Cleaning and Standardization

- Objective: Correct inconsistencies, formatting issues, and data quality problems across relevant columns, such as standardizing NOC country codes and event naming conventions.

SQL Query:

SQL

-- Standardize NOC codes and clean whitespace in athlete names and sports

CREATE OR REPLACE TABLE OLYMPIC_PERFORMANCE_DB.CURATED.CLEAN_ATHLETES AS

SELECT

TRIM(Name) AS Clean_Name,

UPPER(TRIM(Sex)) AS Gender,

Age,

Height,

Weight,

UPPER(TRIM(NOC)) AS Standard_NOC,

INITCAP(TRIM(Sport)) AS Clean_Sport,

INITCAP(TRIM(Event)) AS Clean_Event,

COALESCE(Medal, 'None') AS Standard_Medal

FROM OLYMPIC_PERFORMANCE_DB.RAW.ATHLETE_RAW;

SELECT * FROM OLYMPIC_PERFORMANCE_DB.CURATED.CLEAN_ATHLETES LIMIT 5;

Output / Result:

LEAN_NAME	ENDER	G E	EIGHT	VEIG HT	TANDARD_ NOC	LEAN_SPORT	LEAN_EVENT	TANDARD_MEDAL
Dijiang	M	4	80	0	HN	asketball	asketball Men's Basketball	lone
Lamusi	M	3	70	0	HN	udo	udo Men's Extra-Lightweight	lone
unnar Nielsen	M	4	65	1	EN	thletics	thletics Men's 1,500 metres	lone
dgar Lindenau Aabye	M	4	82	5	EN	ug-Of-War	ug-Of-War Men's Tug-Of-War	old

LEAN_NAME	GENDER	AGE	IEIGHT	VEIGHT	TANDARD_NOC	LEAN_SPORT	LEAN_EVENT	TANDARD_MEDAL
hristine Jacoba Aaftink		1	85	2	IED			

Activity 4: Data Transformation and Validation

- **Objective: Apply advanced data transformations to prepare derived fields (such as BMI or age groups) and build curated summary views for reporting and dashboard integration.**

SQL Query:

SQL

```
-- Create derived analytical table with BMI and Age Groups for modeling
CREATE OR REPLACE TABLE OLYMPIC_PERFORMANCE_DB.MART.ATHLETE_ANALYTICS_MART AS
SELECT
  Clean_Name,
  Gender,
  Standard_NOC,
  Clean_Sport,
  Standard_Medal,
  Age,
  ROUND(Weight / POWER(Height / 100, 2), 2) AS Athlete_BMI,
  CASE
    WHEN Age < 20 THEN 'Teen (Under 20)'
    WHEN Age BETWEEN 20 AND 29 THEN 'Prime (20-29)'
    WHEN Age BETWEEN 30 AND 39 THEN 'Veteran (30-39)'
    ELSE 'Senior (40+)'
  END AS Age_Group
FROM OLYMPIC_PERFORMANCE_DB.CURATED.CLEAN_ATHLETES
WHERE Height IS NOT NULL AND Weight IS NOT NULL;
```

SELECT * FROM OLYMPIC_PERFORMANCE_DB.MART.ATHLETE_ANALYTICS_MART LIMIT 5;

Output / Result:

LEAN_NAME	GENDER	TANDARD_NOC	LEAN_SPORT	TANDARD_MEDAL	AGE	ATHLETE_BMI	AGE_GROUP
Dijiang	M	HN	asketball	lone	4	4.69	rime (20-29)
Lamusi	M	HN	udo	lone	3	0.76	rime (20-29)
unnar Nielsen	M	EN	thletics	lone	4	6.09	rime (20-29)
dgar Lindenau Aabye	M	EN	ug-Of-War	old	4	8.67	eteran (30-39)
hristine Jacoba Aaftink		IED	peed Skating	lone	1	3.94	rime (20-29)

- Validation Note: All transformed tables have been verified for row-level integrity and schema accuracy, confirming they are fully ready for dashboard integration and downstream analytics.